The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the main idea, the keywords, or the summary.
- The topic can be used to organize and structure the information in the text, such as by using headings, subheadings, paragraphs, or bullet points.
- The topic can be used to guide the reader's or listener's attention and interest, as well as to help them understand the purpose and main message of the text.
- The topic can be related to other topics, such as by using examples, comparisons, contrasts, or connections.
- The topic can be analyzed and evaluated, such as by asking questions, making inferences, drawing conclusions, or giving opinions.

B.Sc. (Hons.) in Environmental Science

- B.Sc. (Hons.) in Environmental Science is a three-year undergraduate degree course that covers the study of the natural and human-made environment, its problems and solutions.
- The course aims to provide students with the knowledge and skills to understand, analyze and manage environmental issues and challenges in various domains such as ecology, engineering, conservation, biology and chemistry.
- The course is accredited by the Institution of Environmental Sciences (IES) and follows the Choice Based Credit System (CBCS) as per the National Education Policy-2020.
- The course has 140 credits spread across 26 papers, which include eight electives and four ability-enhancement courses. Two of the courses are skill enhancement papers that help students develop practical and professional skills in environmental science.
- The course covers topics such as environmental planning, disaster management, natural resources management, environmental pollution, environmental impact assessment, environmental law and policy, environmental biotechnology, environmental economics, climate change and sustainability.
- The course also involves field work, laboratory work, projects and internships that expose students to real-world environmental problems and solutions.
- The course prepares students for various career opportunities in the public and private sectors, such as environmental consultants, environmental engineers, environmental educators, environmental researchers, environ-

mental activists, environmental journalists, environmental auditors, environmental managers and environmental policy makers.

Unit 1 - Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness. Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development.

Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness

- **Environment** is the sum total of all the physical, chemical and biological factors and processes that surround and influence a living organism .
- **Types of Environment** are broadly classified into two categories: **natural environment** and **human-made environment**. Natural environment consists of the natural resources and phenomena that exist independently of human intervention, such as air, water, land, climate, plants and animals. Human-made environment consists of the artificial structures and systems that humans create and modify, such as buildings, roads, dams, industries, farms and cities.
- **Components of environment** are mainly divided into two categories: **biotic environment** and **abiotic environment**. Biotic environment includes all living organisms such as animals, birds, forests, insects, reptiles and microorganisms like algae, bacteria, fungus, viruses, etc. Abiotic environment includes all non-living factors and processes such as light, heat, radiation, wind, water, soil, minerals, etc.
- **Segments of environment** are the four important zones that arise from the interaction of the three components of the environment, namely **atmosphere**, **hydrosphere**, **lithosphere** and **biosphere** . Atmosphere is the protective layer of gases that surrounds the earth and sustains life. Hydrosphere is the water component of the earth that covers about 71% of its surface. Lithosphere is the solid crust of the earth that consists of

rocks, minerals and soil. Biosphere is the realm of living organisms and their interactions with the atmosphere, hydrosphere and lithosphere .

- **Scope and importance of environment** are related to the study and understanding of the natural and human-made environment and their interrelationships. The scope of environment covers various disciplines such as ecology, biology, chemistry, physics, geography, sociology, economics, etc. The importance of environment lies in its role in supporting life, maintaining ecological balance, providing resources, regulating climate, influencing culture and human development, etc .
- **Need for public awareness** is the necessity of educating and informing the public about the environment and its issues, such as pollution, degradation, conservation, management, etc. Public awareness is essential for creating a sense of responsibility, participation and action among the people to protect and improve the environment and to achieve sustainable development .

Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem

- **Ecosystem** is a functional unit of the environment that consists of a community of living organisms and their physical environment, interacting as a system .
- **Types of ecosystem** are based on various criteria such as nature, size, location, etc. Some of the common types of ecosystem are **terrestrial ecosystem** (land-based), **aquatic ecosystem** (water-based), **forest ecosystem** (dominated by trees), **grassland ecosystem** (dominated by grasses), **desert ecosystem** (dry and arid), **marine ecosystem** (saltwater), **freshwater ecosystem** (freshwater), **mountain ecosystem** (high altitude), **polar ecosystem** (cold and icy), etc .
- **Structure of ecosystem** refers to the arrangement and organization of the biotic and abiotic components of the ecosystem. The structure of ecosystem can be described in terms of **biomass**, **productivity**, **diversity**, **trophic levels**, **niche** and **habitat** . Biomass is the total mass of living matter in an ecosystem. Productivity is the rate of production of biomass by the ecosystem. Diversity is the variety of species and genes in an ecosystem. Trophic levels are

Unit 2 - Natural Resources: Introduction, Classification. Water Resources; Availability, sources and Quality Aspects, Water Borne and Water Induced Diseases, Fluoride and Arsenic Problems in Drinking Water. Mineral Resources; Material Cycles; Carbon, Nitrogen and Sulfur cycles. Energy Resources; Conventional and Non conventional Sources of Energy. Forest Resources; Availability, Depletion of Forests, Environment impact of forest depletion on society.

Introduction

- Natural resources are substances or materials from nature that humans use to sustain life.
- These resources are naturally available on Earth without any human action.
- Natural resources have economic, ecological, scientific, aesthetic, and cultural values in human life.
- Some natural resources are necessary for life, whereas others have economic value and contribute to industry.
- Examples of natural resources are air, water, soil, wildlife, forests, minerals, and fossil fuels.

Classification

- Natural resources can be classified based on different criteria, such as origin, renewability, availability, distribution, and development.
- Based on origin, natural resources can be classified into biotic and abiotic resources.
  - Biotic resources are derived from living organisms, such as plants, animals, and microorganisms.
  - Abiotic resources are derived from non-living sources, such as rocks, minerals, air, water, and sunlight.
- Based on renewability, natural resources can be classified into renewable and non-renewable resources.
  - Renewable resources are those that can be replenished or regenerated in a short period of time, such as solar energy, wind energy, biomass, and hydroelectric power.
  - Non-renewable resources are those that cannot be replenished or regenerated in a short period of time, such as fossil fuels, minerals, and nuclear energy.

- Based on availability, natural resources can be classified into inexhaustible and exhaustible resources.
  - Inexhaustible resources are those that are available in unlimited quantities and are not affected by human consumption, such as sunlight, air, and water.
  - Exhaustible resources are those that are available in limited quantities and are affected by human consumption, such as fossil fuels, minerals, and wildlife.
- Based on distribution, natural resources can be classified into ubiquitous and localized resources.
  - Ubiquitous resources are those that are found everywhere on Earth, such as air, water, and sunlight.
  - Localized resources are those that are found only in certain regions or countries, such as oil, coal, and gold.
- Based on development, natural resources can be classified into potential and actual resources.
  - Potential resources are those that are not yet utilized or exploited, but have the possibility of being used in the future, such as geothermal energy, tidal energy, and methane hydrates.
  - Actual resources are those that are already utilized or exploited, and have a known quantity and quality, such as oil, coal, and natural gas.

Water Resources

- Water is one of the most essential and abundant natural resources on Earth.
- Water covers about 71% of the Earth's surface, but only 2.5% of it is fresh water.
- Water is used for various purposes, such as drinking, irrigation, sanitation, industry, recreation, and transportation.
- Water resources can be classified into surface water and groundwater.
  - Surface water is the water that is found on the Earth's surface, such as rivers, lakes, streams, ponds, and oceans.
  - Groundwater is the water that is found below the Earth's surface, such as aquifers, springs, and wells.
- Water resources are affected by various factors, such as availability, quality, and demand.
  - Availability of water depends on the precipitation, evaporation, and runoff patterns in a region.
  - Quality of water depends on the physical, chemical, and biological characteristics of the water, such as temperature, pH, dissolved oxygen, nutrients, and contaminants.
  - Demand of water depends on the population, consumption, and development patterns in a region.
- Water resources are facing various challenges, such as scarcity, pollution, overexploitation, and climate change.

- Scarcity

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

Pollution and their Effects

- Pollution is the introduction of harmful substances or energy into the environment that cause adverse effects on living and non-living things.
- Pollution affects the natural balance and creates adverse conditions for life. It has led to severe environmental issues that require immediate action and resolution.
- The different factors that contribute to pollution can be in the form of substances (solid, liquid, gaseous) or energy (sound, heat etc).
- Pollution can be classified into different types based on the source, nature, and effects of the pollutants. Some of the common types of pollution are:
  - Air pollution: The contamination of the atmosphere by harmful gases, particles, or chemicals that affect the quality of the air and the health of living beings .
  - Water pollution: The contamination of water bodies by harmful substances or microorganisms that affect the quality of the water and the aquatic life.
  - Soil pollution: The contamination of the soil by chemicals, wastes, or pathogens that affect the fertility of the soil and the growth of plants.
  - Noise pollution: The excessive or unwanted sound that causes annoyance, stress, or hearing loss to living beings.
  - Solid waste pollution: The accumulation of solid wastes that are not properly disposed of or recycled, and that pose a threat to the environment and public health.
- The effects of pollution on the environment and living beings are manifold and serious. Some of the effects are:
  - Environment degradation: Pollution affects the natural resources and ecosystems that sustain life on Earth. It can reduce the biodiversity, alter the climate, damage the ozone layer, and cause acid rain .
  - Health problems: Pollution can cause various diseases and disorders in humans and animals, such as respiratory infections, asthma, cancer, allergies, cardiovascular diseases, and reproductive problems .

  - Economic losses: Pollution can reduce the productivity and quality of crops, livestock, fisheries, and forests. It can also increase the costs of health care, environmental remediation, and disaster management .

Public Health Aspects of Environmental Pollution

- Public health is the science and practice of protecting and improving the health of populations through the prevention and control of diseases and the promotion of healthy behaviors.
- Environmental pollution is one of the major threats to public health, as it can cause or exacerbate various health problems in humans and animals .
- The public health aspects of environmental pollution include:
  - Assessing the sources, levels, and trends of environmental pollutants and their exposure and health effects on different populations and groups .
  - Developing and implementing policies, regulations, and standards to prevent, reduce, or eliminate environmental pollution and its health impacts .
  - Educating and empowering the public and stakeholders about the causes, consequences, and solutions of environmental pollution and its health effects .
  - Monitoring and evaluating the effectiveness and impact of the interventions and actions taken to address environmental pollution and its health effects .
  - Conducting and supporting research and innovation to generate new knowledge and technologies to combat environmental pollution and its health effects .

Water Pollution

- Water pollution is the contamination of water bodies by harmful substances or microorganisms that affect the quality of the water and the aquatic life.
- Water pollution can be caused by various sources, such as:
  - Domestic sewage: The wastewater from households and other buildings that contains organic matter, nutrients, pathogens, and chemicals.
  - Industrial effluents: The wastewater from industries that contains toxic metals, acids, bases, solvents, dyes, pesticides, and other chemicals.
  - Agricultural runoff: The water that flows from the fields and carries fertilizers, pesticides, animal wastes, and soil particles.

- Urban runoff: The water that flows from the streets and carries oil, grease, litter, and other pollutants.
- Marine dumping: The disposal of solid wastes, such as plastics, into the oceans and seas.
- Oil spills: The accidental or deliberate release of oil

Unit 4 - Current Environmental Issues of Importance

Global Warming

- Global warming is the increase in the average temperature of the Earth's surface and atmosphere due to the enhanced greenhouse effect.
- The greenhouse effect is the natural process by which certain gases in the atmosphere trap some of the outgoing infrared radiation from the Earth and re-emit it back to the surface, keeping the planet warmer than it would be otherwise.
- The enhanced greenhouse effect is the amplification of the natural greenhouse effect due to the increased concentration of greenhouse gases in the atmosphere as a result of human activities, such as burning fossil fuels, deforestation, agriculture, and industrial processes.
- Some of the main greenhouse gases are carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O), and chlorofluorocarbons (CFCs).
- Global warming has many negative impacts on the environment, such as melting of glaciers and ice caps, sea level rise, extreme weather events, droughts, floods, heat waves, wildfires, loss of biodiversity, ocean acidification, coral bleaching, and spread of diseases.

Greenhouse Effect

- The greenhouse effect is the natural process by which certain gases in the atmosphere trap some of the outgoing infrared radiation from the Earth and re-emit it back to the surface, keeping the planet warmer than it would be otherwise.
- The greenhouse effect is essential for life on Earth, as it maintains a suitable temperature range for living organisms.
- The greenhouse effect is named after the analogy of a greenhouse, a glass structure that allows sunlight to enter but prevents heat from escaping, creating a warm environment inside.
- The Earth's atmosphere acts like a greenhouse, as it is transparent to incoming shortwave radiation from the Sun, but opaque to outgoing longwave radiation from the Earth.
- Some of the main greenhouse gases are water vapor (H2O), carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O), and ozone (O3).
- The greenhouse effect can be quantified by the radiative forcing, which is the difference between the incoming and outgoing radiation at the top of the atmosphere, measured in watts per square meter (W/m2).

- The radiative forcing of a greenhouse gas depends on its concentration, its absorption spectrum, and its lifetime in the atmosphere.

Climate Change

- Climate change is the long-term change in the statistical properties of the Earth's climate system, such as temperature, precipitation, wind, and pressure, over periods of decades to millennia.
- Climate change can be caused by natural factors, such as changes in the Earth's orbit, solar activity, volcanic eruptions, and ocean currents, or by human factors, such as greenhouse gas emissions, land use change, aerosol emissions, and nuclear tests.
- Climate change can be detected by observing the changes in the mean and variability of climate variables, such as global mean surface temperature, sea level, precipitation patterns, and frequency and intensity of extreme events.
- Climate change can be projected by using mathematical models that simulate the interactions and feedbacks among the physical, chemical, and biological components of the climate system, such as the atmosphere, the hydrosphere, the cryosphere, the biosphere, and the lithosphere.
- Climate change can have significant impacts on the natural and human systems, such as ecosystems, agriculture, water resources, health, energy, security, and economy.

Acid Rain

- Acid rain is the precipitation that has a pH lower than 5.6, due to the presence of acidic substances, such as sulfur dioxide (SO2) and nitrogen oxides (NOx), in the atmosphere.
- Acid rain is formed when SO2 and NOx react with water vapor and other chemicals in the air, forming sulfuric acid (H2SO4) and nitric acid (HNO3), which then fall to the ground as rain, snow, fog, or dust.
- The main sources of SO2 and NOx are the combustion of fossil fuels, such as coal, oil, and gas, in power plants, industries, and vehicles, and the smelting of metal ores, such as iron, copper, and zinc.
- Acid rain can have harmful effects on the environment, such as acidification of soil and water, leaching of nutrients and metals, damage to vegetation and crops, corrosion of buildings and monuments, and reduction of visibility and air quality.

Ozone Layer Formation and Depletion

- The ozone layer is the region of the stratosphere, between 15 and 35 km above the Earth's surface, where ozone (O3) molecules are concentrated, forming a protective shield that absorbs most of the harmful ultraviolet (UV) radiation from the Sun.
- The ozone layer is formed by the photodissoci

Unit 5 - Environmental Protection

Environmental Protection Act 1986

- The Environmental Protection Act 1986 is an Act of the Parliament of India that aims to provide for the protection and improvement of the environment .
- The Act empowers the Central Government to establish authorities, issue rules and directions, and take measures to prevent, control and abate environmental pollution .
- The Act covers various aspects of environmental protection, such as standards for emission and discharge of pollutants, hazardous substances management, environmental impact assessment, environmental audit, and penalties for violations .
- The Act is widely considered to have been a response to the Bhopal gas leak, one of the worst industrial disasters in history, that occurred in 1984 .

Initiatives by Non Governmental Organizations (NGO's) for environmental protection

- Non Governmental Organizations (NGOs) are non-profit, citizen-led organizations that are not affiliated with the government and work for various social and environmental causes .

- NGOs play an important role in raising environmental awareness, mobilizing public participation, advocating for policy changes, implementing grassroots projects, and monitoring the environmental performance of governments and corporations .

- Some of the prominent NGOs working for environmental protection are:

  - Greenpeace: An international NGO that campaigns for a green and peaceful world by exposing environmental problems and promoting solutions.
  - World Wide Fund for Nature (WWF): An international NGO that works to conserve nature and reduce the threats to biodiversity and ecosystems.
  - Environmental Foundation for Africa (EFA): A Sierra Leone-based NGO that works to restore and protect the environment in post-conflict countries in Africa.
  - Centre for Science and Environment (CSE): An India-based NGO that conducts research, advocacy and education on environmental issues, such as air and water pollution, climate change, and sustainable development.

Human Population and the Environment: Population growth, Environmental Education, Women Education

- The human population has experienced a period of unprecedented growth, more than tripling in size since 1950. It reached almost 7.8 billion in 2020 and is projected to grow to over 8.5 billion in 2030.
- Population growth and distribution have significant impacts on the environment, such as:
  - Consumption of resources such as land, food, water, air, fossil fuels and minerals.
  - Production of waste products such as air and water pollutants, toxic materials and greenhouse gases.
  - Degradation of natural habitats, biodiversity and ecosystem services .
  - Vulnerability to natural disasters, climate change and infectious diseases.
- Environmental education is a process of learning and teaching about the environment and its interrelationships with human and natural systems .
- Environmental education aims to:
  - Increase environmental awareness and knowledge among people .
  - Develop positive attitudes and values towards the environment .
  - Enhance skills and abilities to solve environmental problems and take action .
  - Promote participation and collaboration among different stakeholders .
- Women education is the provision of formal and informal learning opportunities for girls and women to acquire knowledge, skills and competencies for their personal and social development.
- Women education is important for environmental protection because:
  - Educated women tend to have lower fertility rates, which can help reduce population pressure on the environment .
  - Educated women tend to have more access to health care, family planning and economic opportunities, which can improve their wellbeing and empowerment .
  - Educated women tend to have more awareness and involvement in environmental issues, such as natural resource management, conservation and advocacy .
  - Educated women can act as role models and agents of change for their families and communities .